

OECD NUCLEAR ENERGY AGENCY

**Nuclear Science Committee
Working Party on Scientific Issues of Reactor Systems**

**OECD/NRC Benchmark Based on NUPEC PWR
Sub-channel and Bundle Tests (PSBT)
First Workshop (PSBT-1)**

SUMMARY RECORD

**12-13 April 2010
San Piero a Grado Nuclear Research Group
University of Pisa, Italy**

Jim GULLIFORD Jim.GULLIFORD@oecd.org +33 1 45 24 10 72

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OECD/NEA Nuclear Science Committee
and
Committee On The Safety Of Nuclear Installations

**OECD/NRC Benchmark Based on NUPEC PWR Sub-channel and Bundle Tests (PSBT)
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University of Pisa, Italy

Summary Record

Sponsorship

1. At the Sixth and last workshop for the OECD/NRC Benchmark based on NUPEC BWR Full-size Fine-mesh Bundle Tests (BFBT-6), which was held on April 27-28, 2009 in University Park / State College, PA, USA, a new OECD benchmark based on NUPEC PWR Sub-channel and Bundle Tests (PSBT) (as follow-up benchmark activities of the OECD/NRC BFBT benchmark) was introduced. The presentation of the new benchmark included the benchmark database, specification and schedule.

2. The OECD/NRC PSBT benchmark activities are performed as an international project supported by US NRC and METI (Japan), and endorsed by the OECD/NEA. The benchmark team are organized based on the collaboration between USA and Japan as shown in Figure 1. This approach has been very successful in the OECD/NRC BFBT benchmark, which is being completed. It shall be recognized that METI had sponsored the NUPEC PWR subchannel and bundle test project. JNES on behalf of METI provides the test information and measured data. OECD/NEA provides supporting activities for the international PSBT benchmark project. The US team, headed by Prof. Avramova from the Pennsylvania State University (PSU), co-ordinates the benchmark project in cooperation with US NRC. The activity of the US benchmark team (PSU team) is sponsored by the US NRC.

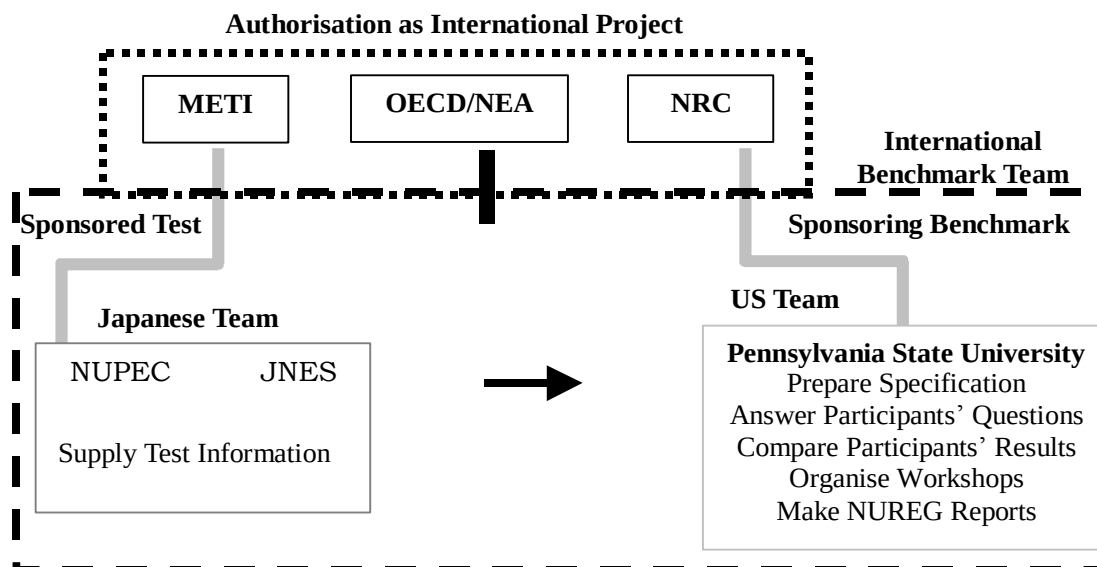


Figure 1. PSBT Benchmark Team

3. This workshop (PSBT-1) was held in conjunction with other meetings, in order to facilitate co-ordination and sharing of work. The three other meetings were held at the same place and during the same week in order to combine efforts in common areas such as CFD modelling and uncertainty analysis and to make the participation more efficient. The meetings concerned were the Fourth Workshop for the OECD Uncertainty Analysis in Modelling (UAM) Light Water Reactor (LWR) benchmark UAM-4 (which took place on April 14-16), and the AER Working Group D workshop (VVER dynamics and safety), which took place on April 15-16, 2010. In parallel with the PSBT-1 meeting also the Second OECD Kalinin-3 VVER-1000 Benchmark Workshop (K3-2) was held at the same premises.

Background and Purpose of the Benchmark Workshop

4. Over the last four years the PSU under the sponsorship of US NRC has prepared, organized, conducted and summarized the OECD/NRC Benchmark based on NUPEC BWR Full-size Fine-mesh Bundle Tests (BFBT). The international benchmark activities have been conducted in cooperation with the Nuclear Energy Agency (NEA), OECD and Japan Nuclear Energy Safety (JNES) organization, Japan. From 1987 to 1995, NUPEC (Nuclear Power Engineering Corporation) in Japan performed a series of void measurement tests using full-size mock-up tests for both BWRs and PWRs. For BWRs, based on the state-of-the-art computer tomography (CT) technology, the void distribution was visualized at a mesh size smaller than the subchannel under actual plant conditions. NUPEC also performed steady-state and transient critical power test series based on equivalent full-size mock-ups. Considering the reliability not only of the measured data, but also other relevant parameters such as the system pressure, inlet sub-cooling and rod surface temperature, these test series supplied the first substantial database for the development of truly mechanistic and consistent models for void distribution and boiling transition.

5. Consequently, the JNES has made available the BWR NUPEC database for the purposes of the OECD/NRC BFBT international benchmark. This international benchmark has encouraged advancement in the uninvestigated fields of two-phase flow theory with very important relevance to the nuclear reactors safety margins evaluation. The BFBT benchmark is made up of two parts (phases), each part consisting of four different exercises and was used for validation of CFD, subchannel, and system codes. The BFBT benchmark activity has been very successful with about thirty (30) organizations from fifteen (15) countries participating in different benchmark exercises. The BFBT activity is being completed with final comparative reports to be published as US NRC NUREG and NEA/OECD reports. A special Nuclear Engineering and Design journal issue devoted to the BFBT benchmark is being prepared and will document the models and results of the participants.

6. Based on the success of the OECD/NRC BFBT benchmark the JNES, Japan has decided to release also the data based on the NUPEC PWR Subchannel and Bundle Tests (PSBT) for an international benchmark and has asked PSU to organize and conduct this benchmark activity. Void fraction measurements and departure from nucleate boiling (DNB) tests were performed at NUPEC under the conditions simulating PWR thermal-hydraulic conditions, including the steady states and the transients such as the power increase, the flow reduction, the depressurization and the temperature increase.

7. The established international OECD/NRC PWR Subchannel and Bundle Tests (PSBT) benchmark, based on the NUPEC database, encourages advancement in subchannel analysis of fluid flow in rod bundles, which has very important relevance to the nuclear reactor safety margin evaluation. The benchmark specification is designed to systematically assess and compare the participants' numerical models for the prediction of detailed subchannel void distributions and DNB to full scale experimental data on a prototypical PWR rod bundle. Currently the numerical modeling of subchannel void distribution has limited theoretical approach that can be applied to a wide range of geometrical and operating conditions. In the past decade, experimental and computational technologies have tremendously improved the study of

the flow structure. Over the next decade, it can be expected that mechanistic approaches will be more widely applied to the complicated fluid phenomena inside fuel bundles. The development of truly mechanistic models for DNB prediction is currently underway. These models must include processes such as void distribution, critical heat flux, and spacer grid behavior. The benchmark specification requires participants to explain their modeling correlations between the measured DNB and the flow dominant processes.

8. It should be recognized that the purpose of this benchmark is not only the comparison of currently available computational approaches but above-all the encouragement to develop novel next-generation approaches that focus on more microscopic processes. In this context, the subchannel grade void fraction data are regarded as the macroscopic data and the computer graphic images are the microscopic data, which provides the detailed void distribution within the subchannel.

9. The test facility design and data from NUPEC includes both macroscopic and microscopic measurements. There are two separate phases, each consisting of different exercises. These phases and exercises are discussed below.

Phase I - Void Distribution Benchmark

10. The purpose of this benchmark phase is three-fold:

1. to provide data for validation of numerical models of void distribution;
2. to provide data over a wide range of geometrical and operating conditions for validation of void distribution models;
3. to provide data for development of mechanistic approaches widely applicable to the fluid phenomena inside fuel bundles.

11. Phase I includes four exercises:

Exercise 1 – Steady-state single subchannel benchmark, which can be used to improve the current models of void generation and void distribution within a subchannel.

Exercise 2 – Steady-state bundle benchmark, which can be applied to subchannel and system thermal-hydraulic codes as well as CFD codes.

Exercise 3 – Transient bundle benchmark, which can be used to determine the capabilities of subchannel and system thermal-hydraulic codes to predict void fraction during transients.

Exercise 4 – Pressure drop benchmark. Code-to-code comparisons will also be performed.

Phase II - Departure from Nucleate Boiling (DNB) Benchmark

12. The purpose of this benchmark is to develop truly mechanistic models for DNB prediction.

13. Phase II includes three exercises:

Exercise 1 – Steady-state fluid temperature benchmark, which can be applied to thermal-hydraulic subchannel and system codes and CFD codes.

Exercise 2 – Steady-state DNB benchmark, which can be used to improve the current models of DNB prediction in PWR bundles at steady state conditions.

Exercise 3 – Transient DNB benchmark, which can be used to improve the current models of DNB prediction in PWR bundles during transients.

Scope and Technical Content of the Benchmark Workshop

14. The technical topics addressed at the workshop included:
- review of the benchmark activities after the 6th OECD/NRC BFBT Workshop;
 - discussion of the PSBT specification;
 - discussion of support studies performed by the benchmark team;
 - discussion of participants' results submitted for Phase I exercises;
 - discussion of final results submitted for BFBT uncertainty analysis exercises;
 - status of the NED special issue with participants' BFBT papers;
 - definition of a work plan and schedule outlining actions to progress on the two phases of the benchmark activities.

Organization and Programme Committee of the Benchmark Workshop

15. An Organization and Programme Committee has made the necessary arrangements for the First Benchmark Workshop, organized the Sessions, and prepared the final program. The general chair was Maria Avramova (PSU). The other members were Hideaki Utsuno (JNES), Stephen M. Bajorek (US NRC), Alexander Velazquez-Lozada (US NRC), José-Maria Aragonés (UPM), representing the NSC, Francesco D'Auria, representing CSNI, Eric Royer (CEA-Saclay), Adrian Tentner (ANL), Kostadin Ivanov (PSU), and Jim Gulliford, representing the OECD/NEA Secretariat.

Opening Session – Introduction and opening remarks

16. The meeting was opened by Prof. P. Terreni, Dean of Engineering Faculty at University of Pisa. He welcomed the participants on behalf of the host organization and wished them a successful work. Dr. C. Parisi gave a quick overview of San Piero a Grado Nuclear Research Group and Facilities and presented their activities and accomplishments. Dr. J. Gulliford presented the structure of working party on scientific issues in reactor systems at the Nuclear Science Committee (NSC) of NEA/OECD, and welcomed the participants on behalf of NEA Secretariat.

17. The agenda was approved with minor adjustments (see *Appendix 1*).

18. The workshop was attended by 48 participants from 28 organizations in 13 countries (see *Appendix 2*). The interest in this benchmark is very large as shown by the participation from research institutions, universities and industry with sub-channel, CFD, porous media and system thermal-hydraulic codes.

Technical Sessions on OECD/NRC PSBT Experimental Database and Final Problem Specifications

19. M. Avramova (PSU) discussed the status of PSBT benchmark activities. The structure and responsibilities of the benchmark team were defined as well as the time frame of the performed and envisioned activities. The following accomplishments were reported:

- a draft of *Volume I: Experimental Database and Final Problem Specifications* was prepared and distributed to the participants (January 2010);
- the benchmark team has tabulated the experimental database;
- the benchmark team has been assisting participants in their modeling efforts through the benchmark e-mail server and web-site (www.nea.fr/html/science/egrs/ltb/PSBT/);
- the benchmark team has provided templates for submission of the participant's results;

- the benchmark team is preparing comparative analysis of the submitted preliminary results for Phase I: Void distribution benchmark.

20. A. Rubin (PSU) made two consecutive presentations on “Discussion of test facilities, measurement methods and test assembly data” and “Discussion of the benchmark phases and exercises and requested output”.

21. In the next presentation, H. Utsuno (JNES) answered several participants’ questions related to the measurement techniques, uncertainties, etc. The following additional clarifications were made:

- CT scanner measurements were conducted by a stepwise rotation within 0.5 mm;
- chordal measurements were already corrected using the CT scanner data. Dr. Utsuno described the void measurement uncertainty and agreed to translate to English and provide to the participants this additional information;
- corrections of the chordal measurements account for the effect of pressure;
- void measurements uncertainties provided in *Volume 1: Experimental Database and Final Problem Specifications* correspond to one standard deviation (1σ);
- Dr. Utsuno suggested to use a two standard deviations (2σ) uncertainty bound in the code-to-data comparisons for the single-measurement tests and a three standard deviations ($3\sigma = \sqrt{(2\sigma)^2 + (2\sigma)^2}$) uncertainty bound for the repeated-measurement tests;
- the height of the beam in the chordal measurements is 1mm;
- the heater rod gap is filled with a non-active gas (such as nitrogen);
- there is no convective heating of the gas;
- the local wall thickness uncertainties are reflected in the specified error in power;
- it was pointed out that it is questionable if the saturated densities of vapor and liquid water were used in the calculation of the void fraction from the measured mixture density.

Technical Sessions on Phase I – Void Distribution Benchmark

22. Sessions 2 to 7 were devoted to the four exercises of Phase I of the PSBT benchmark. The benchmark team summarized, in four (4) presentations, submitted preliminary results for Exercise I-1 (single subchannel steady state void distribution), Exercise I-2 (bundle steady state void distribution), Exercise I-3 (bundle transient void distribution), and Exercise I-4 (pressure drop).

23. Participants made eighteen (18) presentations on the physical models of the computer codes used and the preliminary results obtained for the four exercises of Phase I.

24. Based on these presentations and discussed issues, the following decisions were made:

- Void fraction measurements and associated uncertainties:
 - more detailed explanation of the void fraction (density) measurements will be provided in the final version of the PSBT Experimental Database and Final Problem Specifications;
 - description of the corrections applied to the chordal void fraction measurements and clarification of the reported measurement uncertainties will be provided in the final version of the PSBT Experimental Database and Final Problem Specifications;
 - the additional information given by Dr. Utsuno will be translated to English and distributed to the participants;

- benchmark team will summarize and distribute to the participants a proposal for a modified/corrected measurement uncertainty;
- the time shift in the transient void distribution will be investigated;
- investigation of the deviation between the measured and calculated equilibrium quality for the bundle tests will be performed (M. Valette).
- Exercise cases & boundary conditions:
 - duplicated cases will be identified;
 - the benchmark team will perform sensitivity study on downcomer with CTF and make modeling recommendation in regard to the possible heat transfer between the water flowing through the pressure vessel and the water around the lower part of the bundle (what is the actual fluid temperature at the inlet of the heated length);
 - participants using CFD codes agreed on the following test cases to be modeled for investigation of the axial variation of the void fraction: for the central subchannel configuration S1 of Exercise I-1: 1.2211 (reference case), 1.2223, 1.2237, 1.4326, and 1.4325; and for the bundle configuration B5 of Exercise I-2: 5.2332;
 - code-to-code comparisons of the in-channel void distribution for test sections S3 and S4 (side and corner subchannels) will be included;
 - the bundle average void fractions will be estimated from the available images.
- Spacer grid data:
 - more detailed/accurate geometrical description (*FLECHT SEASET data*) will be obtained;
 - spacer dimensions from the available drawings will be reconstructed;
 - the bundle average loss coefficient will be calculated using the measured overall pressure drop for Bundle B7 and the subchannel loss coefficients will be estimated;
 - a coordinated action among the participants using CFD codes will be taken to provide CAD geometry of the rod bundles.
- Additional requested output to be included:
 - Exercise I-1: Equilibrium quality;
 - Exercise I-1: Images of the in-channel void distribution for test sections S3 and S4 (side and corner subchannels);
 - Exercise I-2: Bundle average void fraction and equilibrium quality.

Conclusions, Actions, and Schedule for conducting the PSBT benchmark

25. The action items and schedule for the next PSBT-2 workshop were discussed. They are provided in the following list.

List of Agreed Actions on PSBT

1. End of May 2010 - Proposal for the modified uncertainty bound will be distributed
2. End of June 2010 - Participants' feedback on the proposed modified uncertainty bound will be collected
3. End of July 2010 - Final version of *Volume 1: Experimental Database and Final Problem Specifications* will be issued
4. End of July 2010 - Templates for the Exercises of Phase II will be distributed
5. End of October 2010 - Preliminary results for Phase II to be submitted
6. End of January 2011 - Final results for Phase I to be submitted

26. The objectives of the next PSBT-2 workshop include:

- discussion of final results of Exercises I-1, I-2, I-3, and I-4;
- discussion on preliminary results of Exercises II-1, II-2, and II-3.

27. The PSBT-2 workshop will be held in conjunction with other meetings, in order to facilitate co-ordination and sharing of work. The three other meetings will be held at the same place and during the same week in order to combine efforts in common areas such as thermal-hydraulic modelling and uncertainty analysis and to make the participation more efficient. The location of the four workshops in April 11-15, 2011 is in Stockholm, Sweden and these workshops will be hosted by KTH. The meetings concerned are the fifth workshop on the OECD Uncertainty Analysis in Modelling (UAM) Light Water Reactor (LWR) benchmark (UAM-5) and AER Working Group D workshop (VVER dynamics and safety), which will take place on April 13-15, 2010. In parallel with the PSBT-2 meeting the third workshop on the Kalinin-3 VVER coupled code benchmark (K3-3) will be held at the same premises.

APPENDIX 1

OECD/NRC Benchmark based on NUPEC PWR Sub-channel and Bundle Tests (PSBT) – First Workshop (PSBT-1)

Hosted by
San Piero a Grado Nuclear Research Group
The University of Pisa, Italy
12-13 April 2010

FINAL AGENDA

Day 1: 12 April 2010

Introduction Session (Joint session with K3-2 workshop) – Chair: A. Petruzzi

- 09:30 – 09:45 Introduction and opening remarks - Dean of Engineering Faculty at University of Pisa
Prof. P. Terreni
- 09:45 – 10:00 A quick overview of San Piero a Grado Nuclear Research Group and Facilities - C. Parisi,
F. D'Auria,
- 10:00 – 10:15 OECD/NEA presentation - J. Gulliford
- 10:15 – 10:30 *Coffee Break*

Session 1 on OECD/NRC PSBT Benchmark Specification – Chair: L. Sabotinov

- 10:30 – 10:45 Overview and status of PSBT benchmark activities – M. Avramova
- 10:45 – 11:15 Discussion of test facilities, measurement methods and test assembly data – A. Rubin, M.
Avramova
- 11:15 – 11:35 Discussion of the benchmark phases and exercises and requested output – A. Rubin, M.
Avramova
- 11:35 – 12:00 Answers to participants' questions – H. Utsuno

Technical Sessions on Phase I – Void Distribution Benchmark

Session 2 Chair: D. Pointer

- 12:00 – 12:25 Summary of comparison and analysis of submitted results for Exercise I-1 – A. Rubin, M.
Avramova
- 12:25 – 12:50 Phase I- Exercise 1, Preliminary simulations with NEPTUNE_CFD – M.C. Galassi
- 12:50 – 13:15 First CFD calculations on PSBT I-1 – E. Krepper, R. Rzehak
- 13:15 – 14:30 *Lunch*

Session 3 Chair: M. Glueck

- 14:30 – 14:55 Preliminary results of Exercise-1 obtained by using STAR-CD – V. Petrov, A. Manera.
- 14:55 – 15:25 Preliminary results for the steady-state sub-channel benchmarks with NEPTUNE_CFD 1.0.8 – C. Baudry, M. Guingo, S. Mimouni, J. Lavieville, A. Douce
- 15:25 – 15:50 Exercise I-1 with F-COBRA-TF – H. Gabriel, M. Gluck
- 15:50 – 16:15 CATHARE2 prediction of void fraction in sub-channels of PSBT benchmark – L. Sabotinov
- 16:15 – 16:30 *Coffee break*

Session 4 Chair: H. Utsuno

- 16:30 – 16:55 Summary of comparison and analysis of submitted results for Exercise I-2 – A. Rubin, M. Avramova
- 16:55 – 17:20 Preliminary results of UNIPI contribution to Exercise 1 and 2 with TRACE5.0, RELAP5.33 and RELAP5 3D – F. Fiori, A. Petruzzi, D. Novog
- 17:20 – 17:55 Steady state single subchannel and bundle simulations by SUBCHANFLOW and TRACE – U. Imke, V. Sanchez Espinoza
- 17:55 – 18:20 Exercise I-2 with F-COBRA-TF – H. Gabriel, M. Gluck

20:00 - *Dinner*

Day 2: 13 April 2010

Session 5 Chair: M. Avramova

- 09:00 – 09:25 Summary of comparison and analysis of submitted results for Exercise I-3 – A. Rubin, M. Avramova
- 09:25 – 09:40 Summary of comparison and analysis of submitted results for Exercise I-4 – A. Rubin, M. Avramova
- 09:40 – 10:05 PSBT Phase 1 results with ASSERT-PV – K. Leung, D.R. Novog
- 10:05 – 10:30 First analyses of PWR Bundle Tests (PSBT) using FLICA-OVAP Thermal-hydraulic core code - Exercise I-2 and I-3", M. Bucci, P.Fillion, A. Kumbaro, A. Frisani
- 10:30 – 10:45 *Coffee Break*

Session 6 Chair: D. Novog

- 10:45 – 11:10 Exercise I-3 with F-COBRA-TF – H. Gabriel, M. Gluck
- 11:10 – 11:35 Preliminary results of UNIPI contribution to Exercise 3 and 4 with TRACE5.0, RELAP5.33 and RELAP5 3D
- 11:35 – 12:00 Four preliminary results of PSBT benchmark, Phase I – M. Valette
- 12:00 – 12:25 Assessment of CATHARE2 code against PSBT experimental data – A. Del Nevo, F. D'Auria
- 12:25 – 12:50 Analysis of PSBT Phase-I Exercises using a Subchannel Code MATRA – Dae-Hyun Hwang
- 13:00 – 14:30 *Lunch*

Session 7 Chair: A. Velazquez-Lozada

- 14:30 – 14:55 Physical models of 3D thermal-hydraulic THYC-COEUR code – Franck Maurel
- 14:55 – 15:20 Update on development of two-phase boiling models using the STAR-CD extended boiling framework – W.D. Pointer, A. Tentner, S. Lo, A. Splawski
- 15:20 – 15:35 Action items and schedule of PSBT benchmark activities – M. Avramova
- 15:35 – 15:55 Update on the completion of the BFBT benchmark publications – K. Ivanov
- 16:00 – 16:15 Discussion of final results on uncertainty analysis exercises (I-4 and II-3) of the BFBT benchmark – M. Avramova
- 16:15 – 16:30 Conclusions and closing remarks – C. Parisi, J. Gulliford

APPENDIX 2

OECD/NEA Workshops in Pisa / Italy

OECD/NRC PSBT Benchmark – First Workshop (PSBT-1)

Host Organization
San Piero a Grado Nuclear Research Group
University of Pisa, Italy
12-13 April 2010

List of Participants

CANADA

BALL, Matt
McMaster University
Department of Engineering Physics
JHE A319 1280 Main St. West, Hamilton, ON, L8S 4K1,
CANADA

Tel: +1 905 962 0246
Fax:
Eml : ballmr@mcmaster.ca

LEUNG, Kenneth
McMaster University
Department of Engineering Physics
JHE A315 1280 Main St. West, Hamilton, ON, L8S 4K1,
CANADA

Tel: +1 905 962 0246
Fax:
Eml: leungk4@mcmaster.ca

KRAUSE, Matt

Atomic Energy of Canada Ltd. (AECL)
Chalk River Laboratory
CANADA, KOJ IJO

Tel: +1 613 584 8811 ext.
43786
Fax:
Eml : krausem@aecl.ca

NOVOG, David R.
McMaster University
1280 Main St. West
Hamilton, Ontario, CANADA, L8S 4K1

Tel: +1 905 525 9140
Fax:
Eml : novog@mcmaster.ca

FRANCE

BAUDRY, Cyril
EDF / R&D
6, Quai Watier
78401 Chatou Cedex France

Tel: +33 130 87 9046
Fax: +33 130 87 7916
Eml: cyril.baudry@edf.fr

BUCCI, Matteo
CEA Saclay
91191 Gif sur Yvette Cedex, France

Tel: +33 1 69 08 84 24
Fax: +33 1 69 08 85 68
Eml: matteo.bucci@cea.fr

FRANCE (cont'd)

DOUCE, Alexandre
EDF / R&D
6, Quai Watier
78401 Chatou Cedex France

Tel: +33 1 30 87 73 08
Fax: +33 1 30 87 79 16
Eml: alexandre.douce@edf.fr

FILLION, Philippe
CEA Saclay
91191 Gif sur Yvette Cedex, France

Tel: +33 0 1 69 08 9140
Fax: +33 0 1 69 08 85 68
Eml: philippe.fillion@cea.fr

GUINGO, Mathieu
EDF / R&D
6, Quai Watier
78401 Chatou Cedex France

Tel: +33 1 30 87 83 72
Fax: +33 1 30 87 79 16
Eml: mathieu.guingo@edf.fr

MAUREL, Franck
EDF / R&D
6, Quai Watier
78401 Chatou Cedex France

Tel: +33 1 30 87 82 05
Fax: +33 1 30 87 77 27
Eml: franck.maurel@edf.fr

PETURAUD, Pierre
EDF / R&D
6, Quai Watier
78401 Chatou Cedex France

Tel: +33 1 30 87 76 04
Fax: +33 1 30 87 77 27
Eml: pierre.peturaud@edf.fr

SABOTINOV, Luben
Institut de Radioprotection et de Sureté
Nucléaire
IRSN DSR/ST3C/BATH

Tel: +33 1 58 35 71 59
Fax: +33 1 58 35 96 02
Eml: luben.sabotinov@irsn.fr

VALETTE, Michel
CEA Grenoble
17 rue des Martyrs 38054
GRENOBLE CEDEX France

Tel: +33 04 38 78 5 5 37
Fax: +33 04 3878 9453
Eml: michel.valette@cea.fr

GERMANY

FRANK, Thomas
ANSYS Germany GmbH
Standenfeldweg 12
D-83624 Otterfing, Germany

Tel: +49 8024 905 476
Fax: +49 8024 9054 17
Eml: thomas.frank@ansys.com

GABRIEL, Harry
AREVA NP
P.O. box 3220
D - 91001 Erlangen (Germany)

Tel: +49 9131 9009 5713
Fax: +49 9131 9004 082
Eml: harry.gabriel@areva.com

GERMANY (cont'd)

GLÜCK, Markus
AREVA NP
P.O. box 3220
D - 91001 Erlangen (Germany)
Tel: +49 9131 900 95694
Fax: +49 9131 900 94082
Eml: markus.glueck@areva.com

IMKE, Uwe
Karlsruhe Institute of Technology (KIT)
Hermann-von-Helmholtz-Platz 1,
Bldg. 521 - 76344 Eggenstein-Leopoldshafen
Germany
Tel: +49 7247 82-3802
Fax: +49 7247 82 3718
Eml: uwe.imke@kit.edu

KREPPER, Eckhard
Forschungszentrum Dresden-Rossendorf
POB 510119 D-01314 Dresden
Tel: +49 351 260 2067
Fax: +49 351 260 2383
Eml: e.krepper@fzd.de

PELLACANI, Filippo
Ntech Department of Nuclear Engineering
Technische Universität
München
Tel: +49 089 2891 5624
Fax: +49 089 2891 5622
Eml: pellacani@ntech.mw.tum.de

SCHEUERER, Martina
Gesellschaft für Anlagen- und
Reaktorsicherheit (GRS) mbH
Forschungsinstitute
D - 85748 Garching, Germany
Tel: +49 89 32004 401
Fax: +49 89 32004 599
Eml: martina.scheuerer@grs.de

WEIS, Johannes
Gesellschaft für Anlagen- und
Reaktorsicherheit (GRS) mbH
Forschungsinstitute
D-85748 Garching, Germany
Tel: +49 89 32004 392
Fax: +49 89 32004 10392
Eml: johannes.weis@grs.de

HUNGARY

PANKA, István
KFKI Atomic Energy Research Institute
H-1525 Budapest 114, P.O.B. 49
Tel: +36 1 3 922 222 ext. 1386
Fax: +36 1 3 959 293
Eml: panka@aeki.kfki.hu

ITALY

DEL FRATE, Licia
Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122
Pisa, Italy
Tel: +39 338 4735111
Fax: +39 050 2210 384
Eml: licia.delfrate@gmail.com

ITALY (cont'd)*DEL NEVO, Alessandro*

Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122 Pisa,
Italy

Tel: +39050 2210 360

Fax: +39 050 2210384

Eml: a.delnevo@ing.unipi.it

FIORI, Filippo

Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122 Pisa,
Italy

Tel: +39 328 9290174

Fax: +39 050 2210 384

Eml: filippofiori@alice.it

GALASSI, Maria Cristina

Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122 Pisa,
Italy

Tel: +39 050 2210361

Fax: +39 050 2210355

Eml: mc.galassi@ing.unipi.it

KOVTONYUK, Andriy

Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122 Pisa,
Italy

Tel: +39 050 2210392

Fax: +39 050 2210 384

Eml: a.kovtonyuk@ing.unipi.it

PETRUZZI, Alessandro

Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122 Pisa,
Italy

Tel: +39 050 2210377

Fax: +39 050 2210 384

Eml: a.petruzzi@ing.unipi.it

ROZZIA, Davide

Gruppo di Ricerca Nucleare San Piero a Grado,
Università di Pisa (GRNSPG/UNIPi)
via Diotisalvi, 2 - c/o prof. D'Auria 56122 Pisa,
Italy

Tel: +39 050 2210 377

Fax: +39 050 2210 384

Eml: daviderozzia@libero.it

JAPAN*UTSUNO, Hideaki*

Japan Nuclear Energy Safety Organization
(JNES) – TOKYUREIT Toranomon Bldg., 3-17-
1, Toranomon, Minato-ku, Tokyo, 105-0001,
Japan

Tel: +81 3 4511 1814

Fax: +81 3 4511 1898

Eml: utsuno-hideaki@jnes.go.jp

KOREA*HWANG, Dae-Hyun*

Korea Atomic Energy Research Institute
(KAERI)

Tel: +82 42 868 8370

Fax: +82 42 868 8990

Eml: dhhwang@kaeri.re.kr

RUSSIAN FEDERATION

PINEGIN, Anatoly
RRC "Kurchatov Institute"
123182, Kurchatov Sq. 1, Moscow, Russia

Tel: +49 351 26 023 18
Fax: +7 499 196 61 72
Eml: pin@vver.kiae.ru

SPAIN

JIMENEZ ESCALANTE, Javier
Universidad Politecnica de Madrid
Dp.to Ingenieria Nuclear
ETSI Industriales Madrid
c\ Josè Gutierrez Aleascal, 2
cp. 28006 Madrid, Spain

Tel: +39 91336 3108
Fax: +39 91336 3002
Eml: jimenez@din.upm.es

SWITZERLAND

PETROV, Victor
Paul Scherrer Institute, Switzerland
5232 Villigen PSI, Switzerland

Tel: +41 56 3102111
Fax: +41 56 3102327
Eml: victor.petrov@psi.ch

SWEDEN

HERNANDEZ-SOLIS, Augusto
Chalmers University of Technology
Fysikgarden 4, 41796 Goteborg, Sweden

Tel: +46 76 2132 403
Fax: +46 31 772 3079
Eml: augusto@nephy.chalmers.se

VINAL, Paolo
Chalmers University of Technology
SE-41296 Goteborg, Sweden

Tel:
Fax: +46 31 772 3079
Eml: paolo@nephy.chalmers.se

UNITED KINGDOM

LO, Simon
CD-adapco
Trident House, Basil Hill Road
Didcot OX11 7HJ

Tel: +44 1235 515333
Fax: +44 1235 515339
Eml: simon.lo@uk.cd-adapco.com

UNITED STATES OF AMERICA

AVRAMOVA, Maria N.
The Pennsylvania State University, USA
231 Reber Building University Park, Pa 16802
USA

Tel: +1 814 865 0043
Fax: +1 814 865 8499
Eml: mna109@psu.edu

UNITED STATES OF AMERICA (cont'd)

BAGLIETTO, Emilio
 CD-adapco, New York
 60 Broadhollow road
 Mewille, NY 11747 USA

Tel: +1 631 549 2300 ext 170
 Fax:
 Eml: emilio@us.cd-adapco.com

IVANOV, Kostadin N.
 The Pennsylvania State University, USA
 231 Reber Building University Park, Pa 16802
 USA

Tel: +1 814 865 0040
 Fax: +1 814 865 8499
 Eml: kni1@psu.edu

KAMEROW, Sophia N.
 The Pennsylvania State University, USA
 231 Reber Building University Park, Pa 16802
 USA

Tel: +1 814 865 2936
 Fax: +1 814 865 8499
 Eml: sln107@psu.edu

POINTER, William D.
 Argonne National Laboratory
 9700 S Cass Ave, NE-208, Argonne, IL USA
 60439

Tel: +1 630 252 1052
 Fax: +1 630 252 4500
 Eml: dpointer@anl.gov

POPOV, Emilian
 ORNL
 1 Bethel Valley rd, Oak Ridge, TN 37831

Tel: +1 865 574 6515
 Fax:
 Eml: popove@ornl.gov

RUBIN, Adam
 The Pennsylvania State University, USA
 231 Reber Building University Park, Pa 16802
 USA

Tel: +1 814 865 2936
 Fax: +1 814 865 8499
 Eml: ajr5052@psu.edu

SUNG, Yixing
 Westinghouse Electric Company
 P.O. Box 355 Pittsburgh, PA 15230-0355 USA

Tel: +1 412 374 2105
 Fax: +1 412 374 2045
 Eml: sungy@westinghouse.com

VELAZQUEZ-LOZADA, Alexander
 U.S. Nuclear Regulatory Commission
 Washington, DC 20555-0001"

Tel: +1 301 251 7509
 Fax: +1 301 251 7423
 Eml: alexander.velazquez-lozada@nrc.gov

OECD

CHOI, Yong-Joon
 OECD NEA
 Le Seine St-Germain 12 bld des Iles
 92123 Issy-Les-Moulineaux, France

Tel: -
 Fax: -
 Eml: yongjoon.choi@oecd.org

GULLIFORD, Jim
 OECD NEA
 Le Seine St-Germain 12 bld des Iles
 92123 Issy-Les-Moulineaux, France

Tel: +33 (0) 145 24 1072
 Fax:
 Eml: jim.gulliford@oecd.org